Power BI Assignment 1

Data Transformation and Data Modeling

Explanation, summary, and submission notes

Prepared for Jason | 5 November 2026

Summary of the saved file

The PBIX stores five tables and 50 measures, plus one blank report page

5

named model tables

50

named measures

1

report page, with no visuals

The submitted copy does not show query steps, DAX definitions, or report visuals. Those items should not be described as verified work.

Source: supplied PBIX package metadata

What the file lets us verify

VERIFIED FROM THE PBIX PACKAGE
One report page named Page 1, sized 1920 by 1080
Five table names and 50 measure names in the All tables layout
No visual-definition files and no separate DataMashup member

NOT VERIFIED FROM THIS COPY
Power Query Applied Steps or M expressions
Column definitions, keys, relationship cardinality, and filter direction
DAX formulas, calculation results, or any specific cleanup operation

Absence of a step list in this package does not prove that no work happened elsewhere. It means this copy cannot substantiate those steps.

Source: supplied PBIX package metadata

Report page evidence

Page 1 1920 × 1080

No report visuals are stored in the page definition

Canvas reconstructed from the embedded page metadata. It is not a direct screen capture.

Model tables

Orders Data	Likely order-level sales data
Order Details	Likely line-item detail
List of Orders	Likely order identifiers or order list
Sales target	Likely target values
Category	Likely product or sales classification

Roles are inferred from names. Keys, columns, and relationship paths are not confirmed here.

Source: supplied PBIX package metadata

Measure inventory

50 NAMED MEASURES

Amount by category or sub-category	10
Profit Margin% by category or sub-category	9
Profit% by category or sub-category	10
Target Amount by category and date grain	20
Order ID Count	1

AGGREGATION PREFIXES

SUM	total
AVG	arithmetic average
MAX	highest value
MEDIAN	midpoint
MIN	lowest value

Counts reflect the measure names saved in the layout. They do not confirm the underlying DAX or data results.

Source: supplied PBIX package metadata

Amount and order-count measures

AMOUNT MEASURES
Five named statistics appear for Category: SUM, AVG, MAX, MEDIAN, MIN
The same five appear for Sub-Category

HOW TO EXPLAIN THE PREFIXES
SUM reports a total when the measure is additive
AVG reports an arithmetic average unless the DAX defines a different calculation
MAX and MIN show the highest and lowest values
MEDIAN shows the middle value in the ordered observations

ORDER ID COUNT
One measure named Order ID Count appears. Its distinct-count or row-count behavior is not visible from the name alone.

Source: supplied PBIX package metadata

Profit percentage measures

PROFIT MARGIN%
By Category: AVG, MAX, MEDIAN, MIN
By Sub-Category: AVG, MAX, MEDIAN, MIN, SUM

PROFIT%
By Category and Sub-Category: AVG, MAX, MEDIAN, MIN, SUM

INTERPRET WITH CARE
A percentage is usually a ratio. Summing row percentages can produce a misleading total, and an unweighted average can differ from the overall margin. Verify each DAX expression and its weighting before interpreting these measures.

Source: supplied PBIX package metadata

Sales target measures and time grain

TARGET AMOUNT BY CATEGORY
SUM, AVG, MAX, MEDIAN, MIN

TARGET AMOUNT BY ORDER DATE
The same five prefixes appear by Year, Month, and Day of Order Date

MODELING NOTE
Time-grain comparisons are only reliable when the target data and order data align at the same grain. The available metadata does not expose date-table design, relationships, or formulas, so those should be checked before describing trends.

Source: supplied PBIX package metadata

Data transformation evidence

WHAT THIS COPY SHOWS

No separate DataMashup member or readable Applied Steps list appears in the archive. The embedded DataModel is XPress9-compressed, so this inspection could not verify its contents.

WHAT THAT MEANS

The transformation sequence is not verifiable from this copy. This does not prove the source data was never transformed; preparation may have happened upstream or in another file.

AVOID CLAIMING WITHOUT EVIDENCE

Data-type changes, null or duplicate handling, filters, merges, appended tables, or calculated columns.

TO DOCUMENT IT

Open Power Query Editor and capture the query name, Applied Steps, key data-type changes, and the resulting columns.

Source: supplied PBIX package metadata

Data modeling notes

WHAT CAN BE SAID
The saved layout names five tables and includes a broad set of measures for amounts, profit percentages, and sales targets.

WHAT TO VALIDATE IN POWER BI
Confirm the grain of each table and the key columns
Check dimension-key uniqueness and relationship cardinality
Verify cross-filter direction and active relationships
Confirm a date table supports day, month, and year analysis
Test each measure against a small manual example

The readable model-layout metadata does not establish the relationships or formulas, so avoid calling this a star schema until the diagram is checked.

Source: supplied PBIX package metadata

Screenshot evidence status

INCLUDED HERE

Blank report-page reconstruction
Five-table name inventory
Measure-family summary
Captions identify metadata-based views

TO ADD IF REQUIRED

Power Query Editor with Applied Steps
Model view showing real relationships
Report view with completed visuals
Any rubric-required transformation evidence

The PBIX has no report visuals to capture, and Power BI Desktop is unavailable here. The evidence panels are metadata reconstructions, not screenshots.

Presentation summary notes

OPENING
“This assignment file contains a Power BI model layout with five named tables and 50 named measures.”

EXPLANATION
“The saved measures cover amount, profit percentage, and target amount. The target measures are named by category and by day, month, and year of Order Date.”

CAVEAT
“The report page is blank in this copy. The package also does not expose Power Query steps or DAX formulas, so I cannot verify the transformation sequence or calculation logic from this file alone.”

CLOSE
“I would add screenshots of the Applied Steps pane and the Model view, then validate the relationships and formulas before final submission.”

Source: supplied PBIX package metadata

Before final submission

CAPTURE
Power Query Editor with the actual queries and Applied Steps
Model view with tables, keys, relationship directions, and cardinalities
Report view after adding required visuals

VERIFY
Column data types and table grain
The logic behind Order ID Count
The DAX and weighting for percentage measures
The date relationships and target-data grain

EXPLAIN
Separate verified work from inferred table roles
Describe each transformation using its actual step name
Use screenshots from Power BI Desktop if the assignment rubric requires them

Source: supplied PBIX package metadata

Appendix: amount and order count

AMOUNT BY CATEGORY

AVG Amt by Category
Max Amt by Category
Median Amt by Category
Min Amt by Category
SUM Amt by Category

AMOUNT BY SUB-CATEGORY

AVG Amt by Sub-Catgry
Max Amt by Sub-Catgry
Median Amt by Sub-Catgry
Min Amt by Sub-Catgry
SUM Amt by Sub-Catgry

Order ID Count

Model names shown exactly as stored.

Source: supplied PBIX package metadata

Appendix: Profit Margin% measures

AVG Profit Margin% By Catgry
AVG Profit Margin% by Sub-Category
Max Profit Margin% By Category
Max Profit Margin% by Sub-Category
Median Profit Margin% by Category
Median Profit Margin% by Sub-Category
Min Profit Margin% by Category
Min Profit Margin% by Sub-Category
SUM Profit Margin% by Sub-Category

Source: supplied PBIX package metadata

Appendix: Profit% measures

AVG Profit% By Category
AVG Profit% by Sub-Category
Max Profit% by Category
Max Profit% By Sub-Category
Median Profit% by Category
Median Profit% by Sub-Category
Min Profit% by Category
Min Profit% By Sub-Category
SUM Profit% by Category
SUM Profit% by Sub-Category

Source: supplied PBIX package metadata

Appendix: target measures by Category

AVG Target Amt by Category
Max Target Amt by Category
Median Target Amt by Category
Min Target Amt by Category
SUM Target Amt by Category

Source: supplied PBIX package metadata

Appendix: target measures by Order Date

DAY

AVG Target Amt by Day of Order Date
Max Target Amt by Day of Order Date
Median Target Amt by Day of Order Date
Min Target Amt by Day of Order Date
SUM Target Amt by Day of Order date

MONTH

AVG Target Amt by Month of Order Date
Max Target Amt by Month of Order Date
Median Target Amt by Month of Order Date
Min Target Amt by Month of Order Date
SUM Target AMT by Month of Order Date

YEAR

AVG Target Amt by Year of Order Date
Max Target Amt by Year of Order date
Median Target Amt by Year of Order Date
Min Target Amt by Year of Order Date
SUM Target Amt by Year of Order Date

Capitalization and spelling follow the saved model names.

Source: supplied PBIX package metadata

Presenter notes

Slide-by-slide speaking points

These notes follow the 19 slides in the presentation. The caveats preserve what the supplied PBIX does and does not show.

Slide 01

Introduce the assignment and explain that this presentation reviews the supplied PBIX file itself. The evidence descriptions distinguish saved model metadata from information that the file does not expose.

Slide 02

Lead with the three observable counts. Five table names and 50 measure names appear in the saved model layout. The report package defines one page, but it has no visual definitions. The package also does not expose step-by-step Power Query transformations.

Slide 03

Explain that the package contains evidence of model objects and page metadata. It does not expose the data-preparation sequence, model relationships, or formula definitions in the readable parts of the archive. Keep absence of evidence separate from a conclusion that no work occurred.

Slide 04

This is an evidence reconstruction from page.json, not a screenshot from Power BI Desktop. The page metadata says Page 1 and 1920 by 1080. The archive contains no visual-definition folder, so there is no saved report visual to summarize.

Slide 05

The readable layout identifies five table names. The role descriptions are careful inferences from those names, not verified descriptions of columns or contents. No relationship lines are drawn because the readable metadata does not establish relationship endpoints.

Slide 06

The count is based on named measure nodes in the All tables layout. The names group into amount, profit margin percentage, profit percentage, target amount, and order count families. Prefixes suggest the aggregation, but the DAX expressions are not readable in this inspection.

Slide 07

Describe the amount measures as a set of summary statistics across Category and Sub-Category. The prefixes suggest intended functions, but inspect the formulas before giving exact definitions. In particular, the name Order ID Count does not say whether the measure counts rows or distinct identifiers.

Slide 08

Explain the two separate percentage families. The saved names include SUM for Profit Margin% by Sub-Category and for Profit% by both Category and Sub-Category. A sum of percentages may not be meaningful if these are ratios, so formula and weighting checks matter. The DAX is not verified in this copy.

Slide 09

The names show a target-amount family by Category and by three Order Date grains: day, month, and year. This indicates intended time analysis. Do not claim that the target values reconcile with actual sales, or that the date relationships work, until those pieces are checked in Power BI.

Presenter notes

Slide 10

Use a precise caveat. The archive does not expose the query step list, so this presentation does not claim that particular cleaning or shaping operations occurred. If those steps were completed, capture the actual Applied Steps pane and source/output columns in Power BI before submitting.

Slide 11

Present this slide as validation guidance, not a critique of unseen model details. The five names suggest an order and target analysis model, but the readable layout alone cannot establish fact and dimension roles or relationship quality. Use Model view and measure formulas to confirm the actual design.

Slide 12

The user requested screenshots. The source copy contains no report visuals, and Power BI Desktop was not available in this environment. Therefore this deck includes clearly labeled metadata reconstructions instead of claiming they are Power BI screen captures. If screenshots are compulsory under the rubric, add real captures from Power BI before submitting.

Slide 13

Read this as a concise submission talk track. If actual screenshots or formula definitions become available, revise the caveat to match the evidence. Keep the distinction between the file's saved state and any work that may have occurred elsewhere.

Slide 14

Use this as the practical checklist before uploading. The three most important missing evidence items are the real Applied Steps, the actual Model view, and report visuals if the assignment expects them. Validate percentage and count definitions before presenting results.

Slide 15

These are the exact saved names from the model layout. Minor spelling and capitalization variations are preserved. They indicate named measure outputs, not verified formula behavior.

Slide 16

This list preserves the model's saved spelling and capitalization. The Category group includes AVG, MAX, MEDIAN, and MIN names. The Sub-Category group also includes a SUM name. The expressions themselves are not available in this inspection.

Slide 17

The model layout contains ten Profit% measure names. Each of Category and Sub-Category has AVG, MAX, MEDIAN, MIN, and SUM prefixes. Confirm the DAX and denominator before interpreting the percentage values.

Slide 18

These are the five saved target-amount measure names for Category. The file's readable metadata does not expose formula definitions or target-data values.

Slide 19

These fifteen saved names cover five prefixes for each Order Date grain: day, month, and year. Their presence shows intended time-grain analysis, but not whether the date relationships or measures return correct values.